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MAST20030: Differential Equations
Tutorial: Week 7

Question 1

(a) Verify that $y_0 = \dfrac{1}{\sqrt{2}}$, $y_n^{(1)} = \cos nx$ and $y_n^{(2)} = \sin nx$ for $n = 1, 2, 3, \ldots$, are eigenfunctions of the linear operator $-\dfrac{\mathrm{d}}{\mathrm{d}x^2}$ subject to the periodic boundary conditions $y(-\pi) = y(\pi)$ and $y'(-\pi) = y'(\pi)$. In so doing identify the eigenvalue for each eigenfunction.

(b) Verify that the eigenfunctions $y_n^{(2)}$ are orthogonal when using the inner product

$$\langle f, g\rangle = \alpha \int_{-\pi}^{\pi} f^*(x)\, g(x)\, \mathrm{d}x.$$

where $\alpha \in \mathbb{R}$ and $\alpha \neq 0$. What is the value of α required for the eigenfunctions to be orthonormal?

Note: If you have done these computations before then please feel free to perform them for the eigenfunctions of the same operator but now with Neumann boundary conditions $y'(-\pi) = y'(\pi) = 0$.

Question 2

Compute the Fourier series of the following functions:

(a) $f(x) = \sin^2 x$ for $-\pi \leqslant x \leqslant \pi$.

(b) $g(x) = |x|$ for $-1 \leqslant x \leqslant 1$.

Since the periodic extension of g is continuous, the Fourier series of g converges to this periodic extension for all $x \in \mathbb{R}$ (*ie.*, it converges *pointwise*). Use this to evaluate

$$\sum_{m=1}^{\infty} \frac{1}{(2m-1)^2} \qquad \text{and, thence,} \qquad \sum_{n=1}^{\infty} \frac{1}{n^2}.$$

Question 3

Consider a signal represented by the function

$$f(t) = \begin{cases} 0, & -\pi < t < -\frac{\pi}{2} \\ \sin t, & -\frac{\pi}{2} < t < \frac{\pi}{2} \\ 0, & \frac{\pi}{2} < t < \pi \end{cases}$$

with $f(t) = f(t + 2\pi)$.

(a) Sketch $f(t)$ for $-2\pi < t < 2\pi$.

(b) What value will the Fourier series converge to when $t = \dfrac{\pi}{2}$?

(c) Determine the general Fourier series for f.

(d) Evaluate the Fourier series at $t = \dfrac{\pi}{2}$, ensuring that your result matches with your answer to (b).

(e) How do the Fourier coefficients decay for large n? Will the Fourier series converge uniformly? Will it exhibit Gibbs phenomena?